

Indian Institute of Technology, Gandhinagar
EE 226 Semiconductor Devices
Semester 2, 2025-2026
Tutorial -2

Q1. How pure are the following Silicon? Compare in absolute terms as well as also compare with respect to the volume density of Si.

- a. Metallurgical-grade Silicon (2N)
- b. Solar-grade silicon (6N)
- c. Electronic Grade Silicon (9N)

Q2. For Hydrogen Atom

- a. Define and plot potential energy for an electron in Hydrogen atom with respect to the radius. How many states electron is allowed to occupy?
 1. Classical model
 2. Bohr model

Q3. For two hydrogen atoms, plot the potential energy and mention the status of allowed energy states wrt potential energy

- a. Both the atoms are non-interacting
- b. Both the atoms are interacting but not very close
- c. Both the atoms are interacting and very close
- d. In which case, H₂ molecules will be stable (i.e. stable co-valent bond). What is the condition for stable co-valent bond formations? Describe the electrons and atoms nucleus in different cases using kinetic energy and potential energy

Q4. For a Silicon atom:

- a. How many electrons does Silicon has?
- b. Do all of the electrons participate in co-valent bond? Which electrons participate?
- c. Draw the potential energy for the simple 2D silicon lattice with co-valent bonds and show all the electrons positions? Which electrons will be bound to their own nucleus and which electrons are shared with another nucleus?
- d. How many neighbours does each atoms has? Will electrons get influenced by nearest nucleus only?
- e. Imagine how will the potential energy would look in 1D, 2D
- f. How many states will be present in this arrangement. Draw potential energy diagram.
- g. Can we see all these states separately? How are the bands are formed? Will there be any band above conduction band? Is there any band below valence band?
- h. Does these bands and the gap between them change if we change the material? Why?

Q5. Define:

- a. What is a bandgap, electron affinity?
- b. A band diagram is a graph showing allowed and forbidden energy levels for electrons in a material, with energy on the vertical axis. Si has 14 electrons, define the status of each electron at 0 K, in which band they will lie.
- c. What is intrinsic semiconductor and extrinsic semiconductor? What are the different ways to make extrinsic semiconductor?

Q6. For intrinsic semiconductor: Draw simplified band diagram (with V_B , C_B and E_{vac})

- At 0K, are there any free electrons?
- At increased temperature (300K), are there any free electrons?
- What is a hole? Will there be any hole at 0K and 300K?
- Can a hole exist without an electron? Does free hole help in conduction?
- How many bonds are broken to have 1 electron-hole pair (EHP)? What happens to the remaining shared electron in the bond? Is it also free?
- Are the number of electrons/holes values fixed for a given temperature? What about their physical location? Will all the electrons stay exactly at E_c or holes at E_v ?
- Is intrinsic concentration at 300K enough to behave as a conductor? How much is this n_i value compared to all the bonds in Si? How to increase the electron/holes number to increase conductivity?

Q7. For extrinsic semiconductor (n-type):

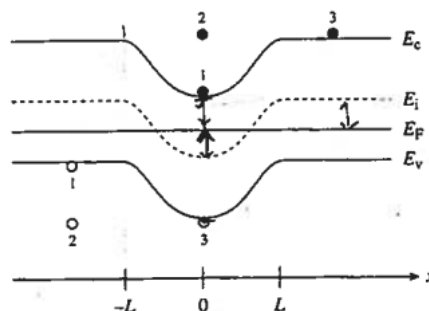
- How does adding pentavalent impurity helps in increasing free electrons? Discuss at 0K and 300K.
- Why would there be a separate energy level in the Silicon bandgap for the impurity atom electron?
- At 300K, how many free electrons will we have? Do we have holes for every electron for the n-type semiconductor?
- Can there be a hole at the donor level?
- Will there be any holes in n-type semiconductor?

Q8. Repeat the same (as in Q7) for p-type semiconductor

Q9. Solid Solubility:

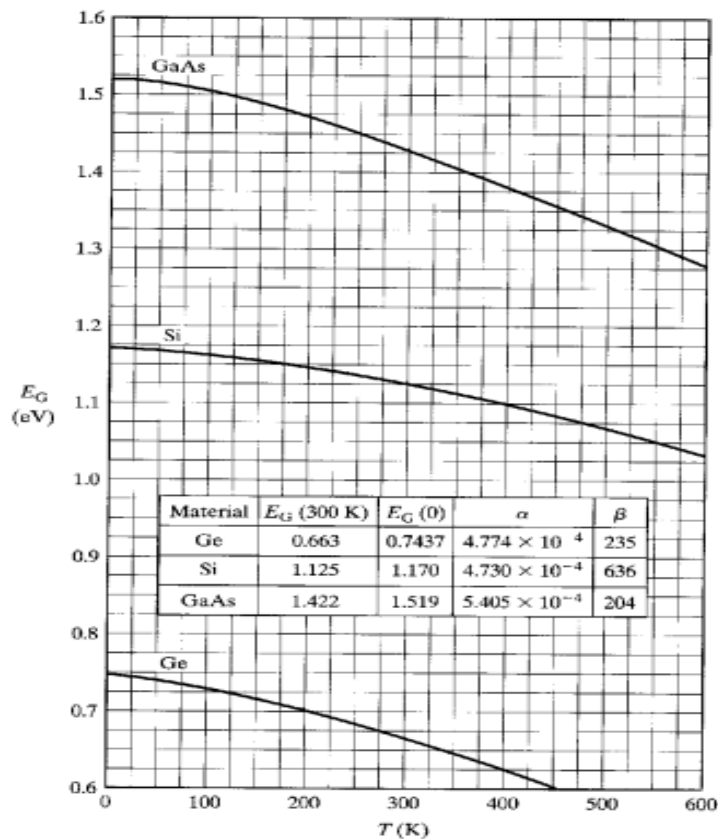
- What is solid solubility?
- Describe any two methods to add impurities in the Si crystal such that they will nicely replace Silicon in the crystal.
Using the solid solubility plot from the lecture notes
- Say 1% impurity is added to Si, which impurity can it be? In absolute value, how much impurity it is?
- What is the maximum doping can be done in Silicon using Phosphorous, Boron and Arsenic and at what temperatures.

Q10. Consider the following energy band diagram. Take the semiconductor represented to be Si maintained at 300 K with $E_i - E_F = E_g/4$ at $x = \pm L$ and $E_F - E_i = E_g/4$ at $x = 0$. Note the choice of E_F as the energy reference level and the identification of carriers at various points on the diagram.



Find the K.E. and P.E. of the electrons and holes pictured on the diagram.

Q11. Read section 3.3.5 on Energy Bands from R. F. Pierret book, and answer the following with proper explanations:



a. At 0 K and 300 K, rank the materials (Ge, Si, GaAs) in order to increase in the band gap using the table in image. Explain the order for both cases.

b. Which material shows the greatest change in band gap over the 0-600 K range?

c. Which material's band gap is least sensitive to temperature?

d. If you were designing a solar cell to operate in hot desert conditions (~350 K), which material among Ge, Si, and GaAs would be most stable in terms of band gap variation, and why?

Q12. Answer the following:

- Compare the intrinsic carrier concentration for Si, Ge and GaAs. What trend can you observe?
- What will be the effect on Melting point of crystal, if lattice constant increases?

Q13.(A) A silicon wafer is uniformly doped p-type with $N_A = 10^{15}/\text{cm}^3$. At $T=0$ K, are there any free holes and electrons?

(B) At $T = 300\text{K}$, assuming all impurities are ionized, are there any free holes and electrons?

Q14. Draw the Energy Band Diagram of two isolated hydrogen nuclei, under the following conditions:

- No energy level is shared.
- Some energy levels are shared.
- Almost all energy levels are shared.

Also, explain under which conditions these situations arise.

Q15. Based on the diagram below, explain how the positioning of E_p affects the strength of covalent bond. [Note E_p is the potential energy maxima between two nuclei, E_{vac} is the vacuum energy level, E_1 , E_2 are the shared energy levels, E_0 is the ground energy level.]

